

Homework 2

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1. For each of the following conics, either find a rational point or prove that there are no rational points:

(a) $x^2 + y^2 = 6$

(b) $3x^2 + 5y^2 = 4$

(c) $3x^2 + 6y^2 = 4$

Proof. (a) If there is a rational solution, we can write it so that the denominators of x and y are equal, which we represent as

$$\left(\frac{a}{c}, \frac{b}{c}\right)$$

for some integers a , b , and c . Then the problem of finding rational solutions to $x^2 + y^2 = 6$ is reduced to the problem of finding integer solutions to $a^2 + b^2 = 6c^2$ where $c \neq 0$.

We can evaluate the right hand side (RHS) mod 9 by noticing that since the only squares mod 9 are 0, 1, 4, and 7, the possible values for $6c^2$ mod 9 are

$$6(0) \equiv 0 \pmod{9}$$

$$6(1) \equiv 6 \pmod{9}$$

$$6(4) \equiv 6 \pmod{9}$$

$$6(7) \equiv 6 \pmod{9}.$$

First, we consider when the RHS is congruent to 0 mod 9. The only pairing of 0, 1, 4, and 7 whose sum is congruent to 0 mod 9 is (0, 0), so we would have

$$a^2 \equiv b^2 \equiv 6c^2 \equiv 0 \pmod{9}.$$

Since (0, 0, 0) is not a solution, at least one of a^2 or b^2 must be nonzero, and thus we could divide by 6 on both sides repeatedly until we no longer had each term being congruent to 0 mod 9. This reduces the problem down to the only remaining case: where the equation is congruent to 6 mod 9. In this case however, we readily see that no pairing of 0, 1, 4, and 7 yields a sum congruent to 6 mod 9, so we conclude that there are no rational solutions to $x^2 + y^2 = 6$.

- (b) Using the same setup as part (a), we can reduce this to the problem of finding integer solutions to $3a^2 + 5b^2 = 4c^2$ where $c \neq 0$. Considered mod 3, we have

$$\begin{aligned} 3a^2 &\equiv 0 \\ 5b^2 &\equiv 0 \text{ or } 2 \\ 4c^2 &\equiv 0 \text{ or } 1. \end{aligned}$$

Clearly, the only combination that works is if $3a^2 \equiv 5b^2 \equiv 4c^2 \equiv 0 \pmod{9}$. But in this case, we run into the same issue as in (a), where we can divide by 3 until we get a term which is not congruent to 0 mod 9, so this equation also has no rational solutions.

- (c) This one is even clearer, since taken mod 3, we get

$$\begin{aligned} 3a^2 &\equiv 0 \\ 6b^2 &\equiv 0 \\ 4c^2 &\equiv 0 \text{ or } 1, \end{aligned}$$

so the only option is $3a^2 \equiv 6b^2 \equiv 4c^2 \equiv 0$, which runs into the same infinite regress argument as before.

□

2. Give a parametrization of the rational solutions to $x^2 + 2y^2 = 3$.

Proof. We start with the point $(1, 1)$, which is a rational solution to the equation, and write out the general form of a line through that point: $y = mx - m + 1$. Intersecting this with $x^2 + 2y^2 = 3$, we get

$$\begin{aligned} 0 &= x^2 + 2(mx - m + 1)^2 - 3 \\ &= x^2 + 2(m^2x^2 + (2m - 2m^2)x + (m^2 - 2m + 1)) - 3 \\ &= x^2 + 2m^2x^2 + (4m - 4m^2)x + 2m^2 - 4m + 2 - 3 \\ &= (2m^2 + 1)x^2 + (4m - 4m^2)x + (2m^2 - 4m - 1) \\ &= (x - 1)((2m^2 + 1)x - (2m^2 - 4m - 1)) \end{aligned}$$

where the third equality comes from using the known root $x = 1$ to simplify the factoring process. Solving for x when $x \neq 1$, then, gives us

$$x = \frac{2m^2 - 4m - 1}{2m^2 + 1}.$$

We can plug this back into the line equation to get

$$\begin{aligned} y &= m \frac{2m^2 - 4m - 1}{2m^2 + 1} - m + 1 \\ &= \frac{2m^3 - 4m^2 - m - (2m^3 + m) + (2m^2 + 1)}{2m^2 + 1} \\ &= \frac{-2m^2 - 2m + 1}{2m^2 + 1}. \end{aligned}$$

Any rational value for m will give us a rational point for x and y , and indeed the line between $(1, 1)$ and any other rational point on the curve will have rational slope, so

$$f(m) = \left\langle \frac{2m^2 - 4m - 1}{2m^2 + 1}, \frac{-2m^2 - 2m + 1}{2m^2 + 1} \right\rangle$$

is a parametrization for the rational solutions to $x^2 + 2y^2 = 3$. □

3. Let C_1 and C_2 be cubics given by the following equations:

$$C_1 : x^3 + 2y^3 - x - 2y = 0 \quad C_2 : 2x^3 - y^3 - 2x + y = 0.$$

Find the nine points of intersection between C_1 and C_2 . One of these is $(0, 0)$ and let $P_1, P_2, \dots, P_8$ be the other 8 points.

Show the linear algebra that implies that if a cubic curve $ax^3 + bx^2y + cxy^2 + dy^3 + ex^2 + fxy + gy^2 + xh + iy + j = 0$ goes through all of $P_1, \dots, P_8$, then $j = 0$. (Which means the curve goes through $(0, 0)$.)

Proof. We begin by setting the curves equal to each other and then consolidating and factoring the x terms on the left side and the y terms on the right side to get

$$x(x+1)(x-1) = 3y(y+1)(y-1).$$

We determine the points of intersection to be $(0, 0)$ along with the other solutions that become obvious upon looking at the above equation:

$$\begin{aligned} P_1 &= (0, -1) \\ P_2 &= (0, 1) \\ P_3 &= (-1, 0) \\ P_4 &= (-1, -1) \\ P_5 &= (-1, 1) \\ P_6 &= (1, 0) \\ P_7 &= (1, -1) \\ P_8 &= (1, 1). \end{aligned}$$

This induces a system of equations on the coefficients of a general cubic curve by plugging in values for x and y , in order, as follows:

$$-d + g - i + j = 0$$

□